

Getting Started with Python

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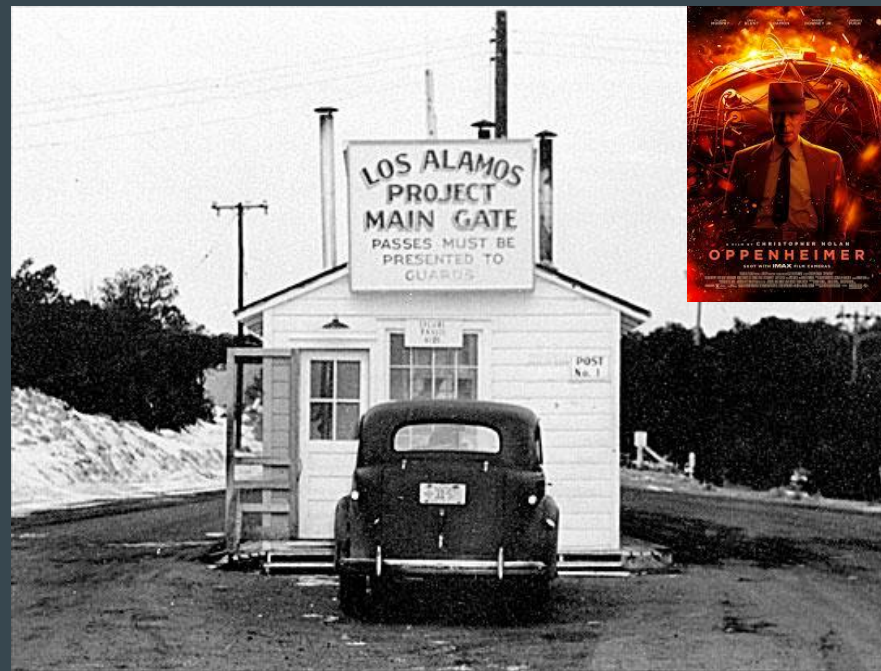
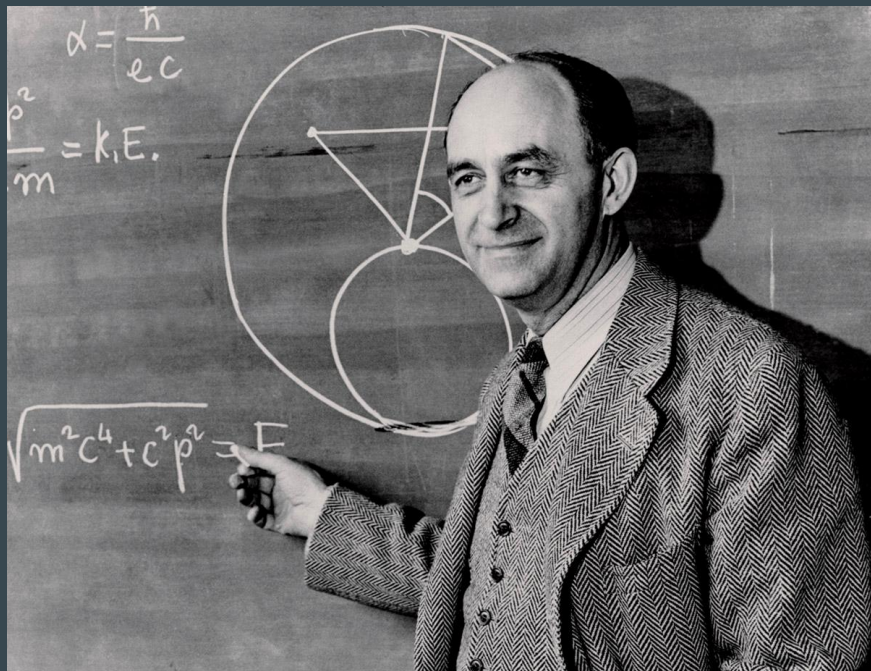
Day 02 - CMSE 801
Prof. Mengsen Zhang

Pre-class Challenges & Questions

- Finding the time and will power to do it.
 - Build a routine?
- Getting survey to work
 - Can copy link to browser – same link for pre-class
- Memorizing syntax
 - You just be able to quickly find it, use it more often like it's a sport
- Order of magnitude approximation (more next)
- Variables and data types (more next)

Fermi Problems/Estimates

Enrico Fermi was an Italian physicist who lived from 1901-1954, worked on the Manhattan project and won the Nobel Prize in Physics.



Fermi Problem

- Modeling example
- Helps you identify the important elements in your model and their interaction/relationships (take it apart and put it back together)
- A “good enough” answer

Example: Drake Equation



Another Example Fermi Estimate

How many piano tuners are there in Chicago?

The classic Fermi problem, generally attributed to Fermi, is "*How many piano tuners are there in Chicago?*" A typical solution to this problem would involve multiplying together a series of estimates that would yield the correct answer if the estimates were correct. For example, we might make the following assumptions:

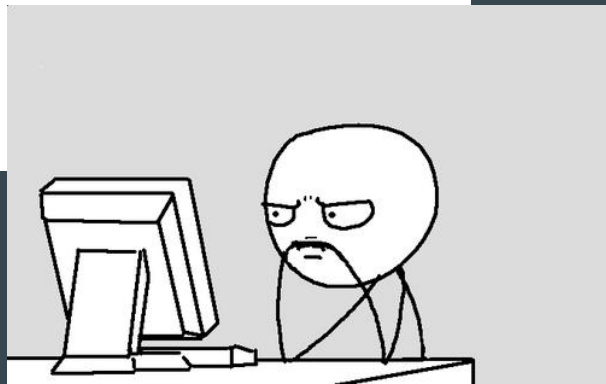
1. There are approximately 5,000,000 people living in Chicago.
2. On average, there are two persons in each household in Chicago.
3. Roughly one household in twenty has a piano that is tuned regularly.
4. Pianos that are tuned regularly are tuned on average about once per year.
5. It takes a piano tuner about two hours to tune a piano, including travel time.
6. Each piano tuner works eight hours in a day, five days in a week, and 50 weeks in a year.

Put that into code

```
1 n_people = 5e6           # No. of people in Chicago
2 ppl_per_household = 2     # 2 persons per house
3 tuner_per_household = 1/20. # 1 tuner every 20 household
4 tune_per_year = 1.0       # Pianos need 1 tune per year
5
6 # No of tunings per year
7 total_tunings = n_people / ppl_per_household * tuner_per_household * tune_per_year
8 print(total_tunings)
9
10 tuning_time = 2           # 2 Hours
11 work_hours = 8            # 8 hours work day
12 workdays = 5             # 5 work days
13 workweeks = 50            # Work weeks in the year
14
15 # No of tunings per year per piano tuner
16 total_tunings_tuner = workweeks * workdays * workhours/tuning_time
17 print(total_tunings_tuner)
18
19
20 # No of tuners in Chicago
21 answer = total_tunings/total_tunings_tuner
22 print(f"There are {answer} tuners in Chicago")
```


Embrace the Error Messages!

```
-----  
NameError                                Traceback (most recent call last)  
<ipython-input-3-501a8a6d760b> in <module>  
    14  
    15 # No of tunings per year per piano tuner  
--> 16 total_tunings_tuner = workweeks * workdays * workhours/tuning_time  
    17 print(total_tunings_tuner)  
    18  
NameError: name 'workhours' is not defined
```



THE GOOD NEWS ABOUT COMPUTERS IS THAT THEY DO WHAT YOU TELL THEM TO DO. THE BAD NEWS IS THAT THEY DO WHAT YOU TELL THEM TO DO.

Today's In-Class Assignment

1. Form Groups of max 4 People
2. Download the Jupyter notebook for Day 02 In-Class Assignment from the website
3. Make sure it is running on your computer, and help the people around you.
4. Assign problem to each group
5. Share out at 9:40 am

Do not forget !!!

- **Submit the survey**
- **Submit Notebook to D2L**
- **Day 03 Pre-Class Due Monday at 5PM**